

Suggested Cultural Practices for Chili Pepper

T.G. Berke, L.L. Black, S.K. Green, R.A. Morris, N.S. Talekar, and J.F. Wang

These suggestions were developed for the specific conditions at the Asian Vegetable Research and Development Center (AVRDC) in the Taiwan lowlands (latitude 23° 7' N, elevation 8 m above sea level). You might need to revise these cultural practices to fit your local environment.

Climate and soil requirements

Chili peppers are better adapted to hot weather than are sweet peppers, but they do not fruit well when night temperatures are high. Optimum temperatures for chili pepper growth range from 20 to 30°C. When the temperature falls below 15°C or exceeds 32°C for extended periods, growth and yield are usually decreased. Peppers are photoperiod-insensitive (day length does not affect flowering or fruit set).

Seed treatment

The primary seed-borne fungal pathogens are surface saprophytes such as *Fusarium* spp. and *Pythium* spp. To minimize seed infection, soak seeds in warm water (50°C) for 30 minutes, rinse them in cold water, and dry them before sowing. Alternatively, dissolve 1 g of Benomyl, 20% active ingredient (AI), wettable powder (WP), and 1 g of Thiram (20% AI WP) [or 0.8 grams of Benlate 50% AI WP] in 400 ml water so that the final concentration is 0.1% AI. Coat the seeds thoroughly by mixing 1 g of seeds with 1 ml of the fungicide mixture.

The primary seed-borne viral pathogens are tobamoviruses, including tobacco mosaic virus (TMV), tomato mosaic virus (ToMV), and pepper mild mottle virus (PMMV). To minimize tobamovirus infection soak seeds in 10% (w/v) trisodium phosphate (TSP) ($\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$) for 30 minutes, transfer them to a fresh 10% TSP solution for 2 hours, then rinse them in running water for 45 minutes. This treatment can be done on freshly harvested or dry seeds. Alternatively, soak seeds for 4–6 hours in 5% (v/v) hydrochloric acid, then rinse in running water for 1 hour.

The primary seed-borne bacterial pathogen is *Xanthomonas campestris* pv. *vesicatoria* (Xcv), the causal agent of bacterial spot. To minimize Xcv infection, soak 2 g of seeds in 10 ml of 1.3% (v/v) acetic acid (shake occasionally) for 4 hours, rinse the seeds with water three times, soak the seeds in 1.25% (v/v) Clorox for 5 minutes, and rinse under running water for 15 minutes. Alternatively, soak seeds in warm water (50°C) for 30 minutes, then dry them or sow immediately. If you sow good quality seeds in sterilized soil, no seed treatment is required.

Seedling production

Average times from sowing to plant emergence at different soil temperatures are shown in Table 1. Germination can vary depending on variety, seed quality, and soil mixture. For optimum germination, sow seeds in well-drained sterile vermiculite at 28°C and water daily.

One gram contains approximately 220 seeds. Approximately 150 g will plant 1 ha at a density of

Table 1. Influence of soil temperature on length of time to plant emergence

Temperature (°C)	Days after sowing
<15	No germination
15	25
20	13
25–35	8
>35	No germination

Source: Knott's Handbook for Vegetable Growers, 1997, p. 96.

26,670 plants/ha, assuming 90% germination and 90% survival after transplanting. Fill the seedling tray with a sowing medium, such as peat moss, commercial potting soil, or a potting mix prepared from soil, compost, rice hulls, vermiculite, peat moss, and/or sand. We recommend a mixture of 70% peat moss and 30% coarse vermiculite. If you use non-sterile materials, we recommend that you sterilize your potting mixture by autoclaving or baking at 150°C for 2 hours. If seedlings are started in a raised soil bed, the soil should be sterilized by burning a 3- to 4-cm thick layer of rice straw or other dry organic matter on the bed. This also adds minor amounts of P and K to the soil, which helps establish the seedlings.

Sow one seed per cell and cover 1 cm deep. Cover the seedlings with an insect-proof net, or sow them inside a greenhouse or screenhouse. This provides shade and protects seedlings from rain and pests, such as aphids, which transmit viruses. Water the tray thoroughly every day, or as needed (not too wet, not too dry), using a fine sprinkler. Leaves should not be wet after nightfall, as this promotes damping off. Irrigate with a 0.25% (w/v) solution of water-soluble fertilizer (10-10-10) when three true leaves appear. Before transplanting seedlings, harden them by exposing them to direct sunlight for 4–5 days; start with 6–8 hours of direct sunlight on the first day and gradually increase the duration each day so that they receive full sun by the 4th day.

Transplanting

Peppers grow best in a loam or silty loam soil with good water-holding capacity and pH within the range 5.5–7.0. Chili peppers should not follow other Solanaceous crops, such as potato or tomato, because these crops share many common soil-borne diseases. Also, because of allelopathic effects, do not plant chili peppers after sweet potatoes.

Recommended spacing varies depending on cropping system, soil type, and variety. At AVRDC, we use 1.5-m wide beds (furrow to furrow), 30 cm high, and we transplant two rows per bed. Rows are 50 cm apart, with 50 cm between plants within rows, for a plant population density of 26,670 plants/ha.

Under favorable growing conditions, the time between sowing to transplanting is four weeks. The ideal seedling has 4–5 true leaves, is disease-free and stocky, and has no flower buds or flowers. Ideally, transplanting should be done in the late afternoon or on a cloudy day, in order to minimize transplanting shock, and the plant should be buried to half its total height (root to tip). The field should be irrigated immediately after transplanting, in order to establish good root-to-soil contact.

Transplanting can be done manually or by machine. If transplanting is done by machine, the transplant trays must be compatible with the transplanter. At AVRDC, we use a Blue Line 2000 mechanical transplanter from Florida. A five-person team can transplant 2500 plants/hour.

Mulching is recommended in order to reduce weed competition and soil compaction, maintain a uniform root environment, conserve soil moisture, and reduce soil erosion. You can use organic mulches, polyethylene plastic, or a combination of both. Plastic mulch must be laid before transplanting; organic mulches can be laid before or after transplanting. If a plastic mulch is used, make holes through which the plants can be transplanted.

Fertilization

Nutrient requirements for a target yield of 5 t/ha (dry matter basis) are listed in Table 2. The actual amount of fertilizer to apply depends on soil fertility, fertilizer recovery rate, soil organic matter, soil mineralization of N, and leaching of N fertilizer. A soil test is recommended to determine the available N, P, and K. The amount of fertilizer to apply can then be calculated based on your target yield and adjusted for residual nutrients. For example, if the target yield is 2.5 t/ha and the soil test indicates that 100 kg each of N, P, and K are available, you would need to apply 125 kg N, 10 kg P, and 100 kg K.

Compost helps to increase soil organic matter content. If substantial leaching of soil nitrate occurs, sidedressing N at monthly intervals is recommended. Ideally, you should conduct your own fertilizer trials

Table 2. N, P, and K requirements, expected recovery rate, and total amount to apply for a target yield of 5 t/ha (dry matter basis) of chili pepper

	Nutrient requirement (kg/ha)	Nutrient recovery (%)	Amount needed* (kg/ha)
N	180	40	450
P	22	10	220
K	200	50	400

* From all sources (soil, fertilizer, crop residues, and compost); actual amount applied should be decreased, based on soil test results and expected nutrient recovery rates.

to determine the optimum fertilizer rate for your area. For information on how to conduct and evaluate fertilizer yield trials, refer to "Determining Fertilizer Rates for Crops in Cropping Systems" in *A Methodology for On-farm Cropping Systems Research*, published in 1981 by the International Rice Research Institute, Los Baños, Philippines. Or, consult your local fertility management specialist for fertilizer recommendations.

Field management

Some varieties require staking to prevent lodging, particularly when they have a heavy fruit load. Yields are generally higher with staking.

Chili peppers have shallow root systems and low tolerance to drought or flooding. Fields should be irrigated if there are signs of wilting at midday. Fields should be drained quickly after heavy rain as peppers will generally wilt and die if they stand in water for more than 48 hours. Practice furrow or drip irrigation; overhead irrigation should be avoided because wet leaves and fruits promote disease. If overhead irrigation must be used, avoid late evening irrigations. Leaves should be dry before nightfall.

If mulch is not available or does not provide adequate weed control, several herbicides are available, such as Lasso (alachlor 43EC), Amex (butralin 47EC), or Dual (metolachlor 8E or 25G). Hand or hoe weeding can be performed thereafter. At AVRDC, we spray 0.4% (v/v) Lasso 43EC around the base of the plants 2–3 days after transplanting, and then spray Roundup

to control weeds in the furrows later in the season. Care must be taken that Roundup does not drift to the pepper plants. The best type of herbicide, rate, and method and date of application will vary depending on, for example, weed species present, soil type, and temperature at time of application.

Harvesting

For fresh use, chili peppers can be harvested either at the green immature or red mature stage. It takes about 55–60 days after flowering for fruits to fully ripen, depending on temperature, soil fertility, and variety. Warmer temperatures will hasten ripening, and cooler temperatures will delay ripening. If conditions are favorable, chili production can continue for several months. Fruits can be harvested weekly as they ripen.

Fresh chili fruits should not be washed unless they will be kept cool (10°C) until sold. Fruits should be stored in a cool, shaded, dry place until they are sold. At typical tropical ambient temperature and humidity (28°C and 60% RH), fruits will last unspoiled for 1–2 weeks. Anthracnose is the major cause of fresh fruit spoilage.

For dry chili, the most important consideration is preservation of the red color of the mature fruits. Drying them in the sun is a common practice, but this tends to bleach the fruits, and rainfall and dew promote fruit rot. Solar dryers have been developed, but they require fairly constant sunshine. Cloudy weather increases the drying time and the risk of post-harvest spoilage. Blanching the fruits in hot water (65°C) for 3 minutes and removing the pedicel and calyx can decrease drying time, increase color retention, and reduce post-harvest losses. In general, varieties with low dry matter content and/or thick flesh are difficult to dry and are generally sold fresh. If ovens are available, dry fruits for 8 hours at 60°C, then reduce the temperature to 50°C and continue until fruits are completely dry (about 10 more hours).

In temperate regions, chili pepper production is usually halted by frost at the end of the season. In tropical and subtropical regions, biotic and/or abiotic stresses (or rotation to a different crop) usually halt chili pepper production.

Table 3. Disease and insect control

Transmission	Appearance	Control methods
Phytophthora blight (crown rot): <i>Phytophthora capsici</i>		
Soil-borne, spread by rain splash. Not transmitted in seeds	Damping off of young seedlings; root and crown rot; blight on leaves, stems and fruits	Resistant varieties; raised beds; crop rotation. Fungicides (RidomilMZ [metalaxyl + mancozeb] and fixed copper
Anthracnose (ripe fruit rot): <i>Colletotrichum capsici</i>; <i>C. acutatum</i>; <i>C. gloeosporioides</i>		
Rain splash, especially in warm wet weather. Can be spread on and in seeds	Mainly affects fruits. Initially, water-soaked spots. Later, spots expand, become dark and depressed, often with concentric rings	Pathogen-free seed; crop rotation; fungicides (Daconil/Bravo 500 [chlorothalonil], Maneb [mancozeb], Zineb
Bacterial wilt: <i>Ralstonia</i> [formerly <i>Pseudomonas</i>] <i>solanacearum</i>		
Promoted by warm weather and poor drainage. Not transmitted in seeds	In late stage leaves turn yellow, wilt and drop. Vascular bundles in lower stem turn brown. Cut stems placed in water ooze milky streams of bacteria from vascular bundles	Resistant varieties; good soil drainage; optimum soil pH 5.5-7.0; rotate with non-Solanaceous crops
Aphid-transmitted viruses: CMV, CVMV, PVY		
Aphids. Not transmitted on seeds	Symptoms are variable	Resistant varieties, weed control, protect young seedlings with nets; IPM
Touch-transmitted viruses: TMV, ToMV, PMMV		
Touch. Also transmitted on and in seeds	Leaf mosaic, plant stunting, systemic chlorosis (bleaching of stems)	Resistant varieties. If not available, use tobamovirus-free seed and dip tools and hands in skim milk before handling plants
Aphids: Cotton aphid (<i>Aphis gossypii</i>) and peach aphid (<i>Myzus persicae</i>)		
	Leaf distortion and cupping. Growth of gray-black sooty mold. Main damage is via viruses	Weed control; crop rotation; IPM. Pesticides (Pirimor [pirimicarb 50DP]; this is safe to predators and parasites of aphids)
Thrips: <i>Scirtothrips dorsalis</i>, <i>Thrips palmi</i>		
	Leaves curl upwards; fruits netted with cork-like streaks	Resistant varieties; weed control; crop rotation; IPM Insecticides (Decis [deltamethrin 2.8EC] Admire/Confidor [imidacloprid 2GR or 10SL], Karate [lambda-cyhalothrin 2.8EC]. Rotate insecticides.
Mites: Broad mite (<i>Polyphagotarsonemus latus</i>)		
	Bronze coloration of growing point and young leaves, older leaves curl downwards, and fruits develop a cork-like distorted surface	Resistant varieties, weed control, crop rotation, IPM. Miticides (sulfur, Avid [abamectin 2EC], Plictran [cyhexatrin 50PW], Kelthane [dicofol 30EC], Omite [propargite 57EC])

IPM = integrated pest management

For more information, please refer to: "Pepper Diseases: A Field Guide", published by AVRDC